EEI6373 Performance Modelling

Mini Project Deliverable - Hospital Outpatient Department (OPD) Queue System

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Introduction / High-Level Problem

The hospitals deal with quite a huge number of patients within a day, and one of the service centers is the Outpatient Department (OPD), which provides consultation to the patients without admitting them in the hospital. Efficient management of arrivals of the patients and doctors' services is essential for maintaining service quality and minimizing waiting time for patients.

The objective of this project is to simulate and analyze the Hospital OPD Queue System. This particular OPD works between 8:00 AM to 4:00 PM, providing consultation to around 300 patients in a working day.

The rate of arrival is not constant throughout the day. There will be more patients expected to arrive between 8:00 AM to 12:00 PM, forming a peak hour. These changes in the arrival rate may cause congestion, with longer waiting times for patients, especially when the arrival rate is greater than the service capacity.

The existence of varying arrival rates, varying priority levels, consultation times, and the availability of doctors makes the system form a queuing system. Hence, the system is an appropriate model for performance analysis.

System flow diagram

Patient Arrival → Join OPD Queue → Priority Assignment → Doctor Assignment → Consultation → Patient Served / Leaves OPD

Performance Objectives

The main objective of this project is to evaluate the performance efficiency of the Hospital OPD Queue System. The following key performance objectives will be considered:

- **Minimize Average Waiting Time:** Reduce the amount of time patients wait before receiving a consultation.
- **Maximize Throughput:** Increase the total number of patients that can be served within the OPD operating period.
- **Identify Performance Bottlenecks:** Identify time periods or resource conditions that result in long queues and increased patient waiting times.
- **Optimize Resource Utilization:** Evaluate the utilization of the available doctors and determine whether the available service resources are effectively used during busy and non-busy periods.

- **Analyze Scalability:** Observe how the OPD queue system behaves when the number of patients or available doctors changes.

Dataset

System Assumptions

- The OPD operates continuously from **8:00 AM to 4:00 PM**.
- Five doctors are available for consultation.
- Patient arrivals are higher during **8:00 AM–12:00 PM**.
- Staff patients have the highest priority, followed by lab-report patients and normal patients.
- Consultation times vary according to patient type.
- Each patient is assigned to one available doctor.

System Overview

Parameter	Description
System	Hospital Outpatient Department (OPD) Queue System
Session Duration	8 AM – 4 PM (8 hours = 480 minutes)
Doctors Available	5 doctors
Total Patients	300
Peak Hours	8 AM – 12 PM
Patient Types	Normal, Staff, Lab-report
Priority Order	Staff > Lab-report > Normal
Service Time (consultation)	Randomly distributed (Normal: 8–20 min, Staff: 5–10 min, Lab-report: 3–8 min)
Performance Variables	Waiting time, Service time, Throughput, Doctor utilization

Dataset Description

Column	Description
patient_id	Unique identifier for each patient
patient_type	Type of patient (normal, staff, lab_report)
arrival_time	Simulated arrival time between 8:00 AM and 4:00 PM
waiting_time	Time in minutes the patient waits before consultation

service_time	Duration in minutes of the doctor's consultation
assign_doctor	Doctor ID assigned to the patient (Dr_1 to Dr_5)

Data Set :
https://drive.google.com/file/d/1yLkkjSTljacwdMeVZDctLWLf4kwWXmkq/view?usp=sharing